

Homework SP Farmer example with price effect

When yields are good for the farmer, they are usually also good for many other farmers. The supply is thus increasing, which will lower the prices. As an example, **we may consider prices going down by 10% for corn and wheat when yields are above average (scenario 1) and going up by 10% when yields are below average (scenario 3).**

1. Formulate the model where these changes in prices affect both sales and purchases of corn and wheat. Assume sugar beet prices are not affected by yields.

Decision variable

x_j = acres of land devoted to crop j , $j = 1, 2, 3$
 w_{11} = tons of wheat sold in scenario $s = 1$
 y_{11} = tons of wheat purchased in scenario $s = 1$
 w_{21} = tons of corn sold in scenario $s = 1$
 y_{21} = tons of corn purchased in scenario $s = 1$
 w_{31} = tons of sugar beets sold at favorable price in sc. $s = 1$
 w_{41} = tons of sugar beets sold at lower price in sc. $s = 1$

 w_{12} = tons of wheat sold in scenario $s = 2$
 y_{12} = tons of wheat purchased in scenario $s = 2$
 w_{22} = tons of corn sold in scenario $s = 2$
 y_{22} = tons of corn purchased in scenario $s = 2$
 w_{32} = tons of sugar beets sold at favorable price in sc. $s = 2$
 w_{42} = tons of sugar beets sold at lower price in sc. $s = 2$

 w_{13} = tons of wheat sold in scenario $s = 3$
 y_{13} = tons of wheat purchased in scenario $s = 3$
 w_{23} = tons of corn sold in scenario $s = 3$
 y_{23} = tons of corn purchased in scenario $s = 3$
 w_{33} = tons of sugar beets sold at favorable price in sc. $s = 3$
 w_{43} = tons of sugar beets sold at lower price in sc. $s = 3$

Objective function

objective fn

$$\begin{aligned}
 \min \quad & 150x_1 + 230x_2 + 260x_3 \\
 & + \left[\frac{1}{3}(1.1) \left(238y_{11} + 290y_{21} - 170w_{11} - 150w_{21} \right) - 36w_{31} - 10w_{41} \right] \\
 & + \left[\frac{1}{3}(1) \left(238y_{12} + 290y_{22} - 170w_{12} - 150w_{22} \right) - 36w_{32} - 10w_{42} \right] \\
 & + \left[\frac{1}{3}(0.9) \left(238y_{13} + 290y_{23} - 170w_{13} - 150w_{23} \right) - 36w_{33} - 10w_{43} \right]
 \end{aligned}$$

Constraint

$x_1 + x_2 + x_3 = 500$			First-stage constraint: Shared across all scenarios (non-anticipativity)		
$3x_1 + y_{11} - w_{11} \geq 200$	$2.5x_1 + y_{12} - w_{12} \geq 200$	$2x_1 + y_{13} - w_{13} \geq 200$			
$3.6x_2 + y_{21} - w_{21} \geq 240$	$3x_2 + y_{22} - w_{22} \geq 240$	$2.4x_2 + y_{23} - w_{23} \geq 240$			
$w_{31} + w_{41} \leq 24x_3$	$w_{32} + w_{42} \leq 20x_3$	$w_{33} + w_{43} \leq 16x_3$			
$w_{31} \leq 6000$	$w_{32} \leq 6000$	$w_{33} \leq 6000$			

2. Solve this model

G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF
	Obj-Sc	150	230	260	238	210	-170	-150	-36	-10	238	210	-170	-150	-36	-10	238	210	-170	-150	-36	-10			
	Prob	1.00	1.00	1.00	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33			
	price mult	1	1	1	0.9	0.9	0.9	0.9	1	1	1	1	1	1	1	1	1.1	1.1	1.1	1.1	1.1	1			
	Obj * price mult	150	230	260	214	189	-153	-135	-36	-10	238	210	-170	-150	-36	-10	262	231	-187	-165	-36	-10			
		x1	x2	x3	y11	y12	w11	w12	w31	w41	y12	y22	w12	w22	w32	w42	y13	y23	w13	w23	w33	w43			
	Variables	138	89	273	0	0	181	53.3	6000	0	0	0	146	26.7	5455	0	0	0	111	0	4909	0			
	Obj	150	230	260	71.4	63	-51	-45	-12	-3.3	79.3	70	-57	-50	-12	-3.3	87.3	77	-62	-55	-12	-3.3	-112406.90		
	Land	1	1	1																			500.00	=	500
Yield Above Avg (Price decrease)	Wheat	2.8	0	0	1	0	-1	0	0	0													200.00	>=	200
	Corn	0	3.3	0	0	1	0	-1	0	0													240.00	>=	240
	BeetSold	0	0	22	0	0	0	0	-1	-1													0.00	>=	0
	BeetFavPrice	0	0	0	0	0	0	0	-1	0													-6000.00	>=	-6000
Yield Avg	Wheat	2.5	0	0							1	0	-1	0	0	0							200.00	>=	200
	Corn	0	3	0							0	1	0	-1	0	0							240.00	>=	240
	BeetSold	0	0	20							0	0	0	0	-1	-1							0.00	>=	0
	BeetFavPrice	0	0	0							0	0	0	0	-1	0							-5454.55	>=	-6000
Yield Below Avg (Price increase)	Wheat	2.3	0	0													1	0	-1	0	0	0	200.00	>=	200
	Corn	0	2.7	0													0	1	0	-1	0	0	240.00	>=	240
	BeetSold	0	0	18													0	0	0	0	-1	-1	0.00	>=	0
	BeetFavPrice	0	0	0													0	0	0	0	-1	0	-4909.09	>=	-6000

	A	B	C	D	E
1		Wheat	Corn	Sugar Beets	
2	Yield (T/acre)	2.5	3	20	
3	Planting cost (\$/acre)	150	230	260	
4	Minimum req (T)	200	240	-	
5	Purchase price (\$/T)	238	210	-	
6				under 6000 T	above 6000 T
7	Selling price (\$/T)	170	150	36	10
8	Profit per acrea (\$/acre)	275	220	460	-60
9	Ranking of profit per acre	Rank #2	Rank #3	Rank #1	Rank #4
10					

Appendix

Solver Parameters

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Set Objective:

SADS8

↑

To:

☐ Max

☒ Min

☐ Value Of:

0

By Changing Variable Cells:

\$I\$7:\$AC\$7

↑

Subject to the Constraints:

SADS9 = \$AF\$9
SADS10:SADS21 >= \$AF\$10:\$AF\$21

▲

▼

Add

Change

Delete

Reset All

Load/Save

☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Simplex LP

▼

Options

Solving Method

Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Help

Solve

Close